Tutorial -1

A Smart farming System predicts whether a plant will grow well (1) or poorly (0) based on daily sunlight and water amount in liters.

Dataset:

Sunlight (x_1) hrs	Water (x_2) ltr	Bais (x_3)	Target (y)
3	1	1	0
8	4	1	1
5	2	1	0
10	5	1	1
7	3	1	1

Given bias $b = 1$.

Initial weights $w_1 = 0, w_2 = 0, w_3 = 0$
$\eta = 0.1$

$$\left.\begin{matrix} w_1=0, w_2=0, w_3=0 \\ \eta = 0.1 \end{matrix}\right\} \text{if } \begin{matrix} z \geq 0 = 1 \\ z < 0 = 0 \end{matrix}$$

Sample One (3,1,1)

$$z = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

$$z = (0)(3) + (0)(1) + (0)(1) + 1$$

$$= 1$$

$$\hat{y} = 1 \qquad \text{Error: } y - \hat{y} \Rightarrow 0 - 1 \Rightarrow -1$$

So Update Weight.

$w_1^{new} = w_1^{old} + \eta e x_1$
$= 0 + (0.1)(-1)(3)$
$= \cancel{0.1} - 0.3$

$w_2^{new} = w_2^{old} + \eta e x_2$
$= 0 + (0.1)(-1)(1)$
$= -0.1$

$w_3^{new} = w_3^{old} + \eta e x_3$
$= 0 + (0.1)(-1)(1)$
$= -0.1$

$b^{new} = b^{old} + \eta e$
$= 1 + (0.1)(-1)$
$= 1 - 0.1$
$= 0.9$

Sample 2 : (8, 4, 1)

$w_1 = -0.3$, $w_2 = -0.1$, $w_3 = -0.1$, $b = 0.9$

$z = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$
$= -0.3(8) + (-0.1)(4) + (-0.1)(1) + 0.9$
$= -2.4 + (-0.4) + (-0.1) + 0.9$
$= -2.4 - 0.4 - 0.1 + 0.9$
$= -2$

$\hat{y} = 0$

Error $y - \hat{y} = 1 - 0 = 1$

Update wts:

$w_1^{new} = w_1^{old} + \eta e x_1$
$= -0.3 + (0.1)(1)(8)$
$= 0.5$

$w_2^{new} = w_2^{old} + \eta e x_2$
$= -0.1 + (0.1)(1)(4)$
$= -0.1 + 0.4$
$= 0.3$

$W_3^{new} = W_1^{old} + \eta e x_3$

$= -0.1 + (0.1)(1)(1)$

$= -0.1 + 0.1$

$= 0$

$b^{new} = b^{old} + \eta e$

$= 0.9 + (0.1)(1)$

$= 1$

Sample 3 (5, 2, 1)

$W_1 = 0.5,\ W_2 = 0.3,\ W_3 = 0,\ b = 1$

$Z = W_1 x_1 + W_2 x_2 + W_3 x_3 + b$

$= 0.5(5) + 0.3(2) + 0(1) + 1$

$= 2.5 + 0.6 + 1 \Rightarrow 4.1$

$\hat{y} = 1$

Error: $y - \hat{y} = 0 - 1 \Rightarrow -1$

Update wts:

$W_1^{new} = W_1^{old} + \eta e x_1$

$= 0.5 + (0.1)(-1)(5)$

$\Rightarrow 0.5 - 0.5$

$= 0$

$W_2^{new} = W_2^{old} + \eta e x_2$

$= 0.3 + (0.1)(-1)(2)$

$= 0.3 - 0.2$

$\Rightarrow 0.1$

$W_3^{new} = W_3^{old} + \eta e x_3$

$= 0 + (0.1)(-1)(1)$

$\Rightarrow -0.1$

$b^{new} = b^{old} + \eta e$

$= 1 + (0.1)(-1)$

$\Rightarrow 1 - 0.1 \Rightarrow 0.9$

Sample 4 $(10, 5, 1, \overset{y}{1})$

$w_1 = 0$, $w_2 = 0.1$, $w_3 = -0.1$, $b = 0.9$

$z = w_1x_1 + w_2x_2 + w_3x_3 + b$

$= 0(10) + 0.1(5) + (-0.1)(1) + 0.9$

$= 0 + 0.5 - 0.1 + 0.9 \Rightarrow 1.3$

$\hat{y} = 1$

Error: $y - \hat{y} = 1 - 1 = 0$

No update wts.

Sample 5 $(7, 3, 1, \overset{y}{1})$

$w_1 = 0$, $w_2 = 0.1$, $w_3 = -0.1$, $b = 0.9$

$z = w_1x_1 + w_2x_2 + w_3x_3 + b$

$= 0(7) + 0.1(3) + (-0.1)(1) + 0.9$

$= 0 + 0.3 - 0.1 + 0.9 \Rightarrow 1.1$

$\hat{y} = 1$

$\therefore$ Error: $y - \hat{y} = 1 - 1 = 0$.

No update wts.

final wt: $w_1 = 0$, $w_2 = 0.1$, $w_3 = -0.1$